QSCP Hamilton Inspection Department

Inspection Report

Site: Hamilton Plant

No.	Status	Date
QSCP-INS-2005-019	Issued	2005-06-13

External Visual Inspection - ATU-100 Piping

Scope and Objectives

This report details the 5-yearly external visual inspection of the piping systems within the ATU-100. The scope encompasses all Class 1 and Class 2 circuits as defined in QSCP-IP-ATU100 Rev 0, focusing on the identification of atmospheric corrosion, insulation degradation, and structural support integrity. The inspection follows the requirements for external examinations to ensure the continued mechanical integrity of the Amine Treating Unit.

Douglas Ferreira serves as the lead inspector for this campaign, holding responsibility for the assessment of piping condition and the verification of compliance with site-specific asset integrity protocols. The inspection activities are executed in accordance with the methodologies established in API 570 and the recommended practices for piping components outlined in API 574.

Reference Data and Inspection History

The baseline for this assessment is established by QSCP-IP-ATU100 Rev 0, which identifies 118 CMLs and sets the class-based intervals for the unit. This document serves as the second periodic external review, following the previous findings documented in QSCP-INS-2000-018. The historical data from the first 5-yearly external visual inspection (Class 1/2) provides the necessary context for evaluating current corrosion trends.

Douglas Ferreira has reviewed the previous records in QSCP-INS-2000-018 to identify areas of concern, specifically focusing on the regenerator overhead and amine circuits. The application of API 570 standards ensures that the transition from the 2000 inspection to the current 2005 cycle maintains continuity in the monitoring of the 118 CMLs defined in the original piping inspection plan.

Inspection Methodology

The external examination was conducted per API 574 guidelines, involving a comprehensive walk-down of the ATU-100 piping. The inspection focused on the condition of the coating systems, the presence of corrosion under insulation, and the state of pipe supports and hangers. All observations are measured against the retirement thickness criteria, where any reading below t-min necessitates the issuance of an NCR.

Remaining life calculations are performed by Douglas Ferreira using the larger of the long-term and short-term corrosion rates. Per site rules, the short-term rate is derived from the latest pair of readings at least 3 years apart. The next thickness reading for Class 1 piping is due at no more than half the remaining life, not exceeding the 5-year maximum interval specified for sour gas and regenerator overhead services.

Responsibilities and Compliance

The QSCP Hamilton Inspection Department is responsible for maintaining the records of the ATU-100 and ensuring that all Class 1, 2, and 3 circuits adhere to the mandated inspection frequencies. Douglas Ferreira is tasked with the final evaluation of the piping condition and the determination of the next inspection date based on the calculated remaining life and the class maxima of 5 years for Class 1 and 2 external visuals.

Adherence to API 570 is mandatory for all pressure piping inspections within the facility. The findings of this 5-yearly external visual inspection are integrated into the asset integrity management system to update the schedules originally set forth in QSCP-IP-ATU100 Rev 0. This ensures that the Amine Treating Unit operates within the safety margins defined by the site retirement thickness and corrosion rate rules.

Sign-off	Name	Function	Date	Signature
Prepared	Douglas Ferreira	Inspection Supervisor, API 570 inspector	2005-06-13	_____